0580_s22_qp_32_Q1

1 Antonio has a shop near the beach.

(a) **(i)** He makes a tally of the number of ice creams he sells on Friday.

~~||||~~ ~~||||~~ ~~||||~~ ~~||||~~ ~~||||~~ ~~||||~~ ~~||||~~ ~~||||~~ ~~||||~~ |||

Work out the number of ice creams he sells on Friday.

..... [1]

(ii) 15 of the ice creams he sells on Friday are vanilla.

Work out the fraction of ice creams he sells on Friday that are vanilla.
Give your answer in its simplest form.

..... [1]

(iii) He buys tubs of ice cream for his shop in the ratio vanilla : chocolate = 11 : 7.
He buys 28 tubs of chocolate ice cream.

Work out how many tubs of vanilla ice cream he buys.

..... [2]

(b) Antonio records the number of chairs his shop hires out on each day for a week.

123 98 116 45 67 165 156

(i) Work out the range.

..... [1]

(ii) Find the median.

..... [2]

(iii) Calculate the mean.

..... [2]

0580_s22_qp_32_Q1

(c) **(i)** Antonio buys beach balls for \$2.50 each and sells them for \$4.20 each.

Work out the percentage profit he makes on each beach ball.

..... % [2]

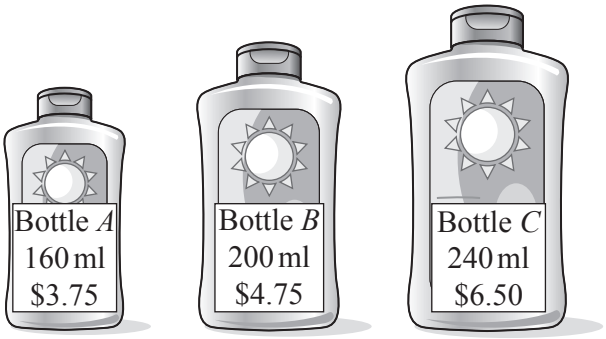
(ii) A beach ball is a sphere with radius 15 cm.

Calculate the volume of the beach ball.
Give the units of your answer.

[The volume, V, of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.]

.... [3]

(d) The shop sells sun cream in bottles A, B and C.



NOT TO SCALE

Work out which bottle is the best value.
You must show all your working.

Bottle ..... [3]